

Sentiment Analysis Using Transformer Models: BERT, T5, and GPT

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Abstract—Sentiment analysis, or opinion mining, is a crucial Natural Language Processing task that classifies text into sentiment categories such as positive, negative, or neutral. While early approaches relied on lexicon-based methods and traditional machine learning models, recent advances in deep learning, particularly transformer-based architectures, have significantly improved sentiment classification performance. This paper investigates the application of state-of-the-art transformer models to sentiment analysis, with a focus on Slovak-language datasets. We evaluate their effectiveness using three datasets: SentiSK, Senti-grade, and the Slovak dataset for SA. Our experiments include both three-class and two-class classification tasks, comparing the performance of RoBERTa, DistilBERT, mT5, byT5, and GPT models based on accuracy, F1-score, precision, and recall. The results demonstrate that GPT and mT5 achieve the highest performance across various datasets, while DistilBERT shows strong potential for specific tasks. Our findings highlight the advantages of transformer-based models for sentiment analysis and emphasize the importance of dataset-specific optimizations.

Keywords—BERT, dataset, GPT, LLM, NLP, sentiment analysis, Slovak language, T5, Transformers

I. INTRODUCTION

SENTIMENT ANALYSIS, often referred to as opinion mining, is a natural language processing (NLP) technique designed to assess the emotional tone of text [1]. Its goal is to classify written content into sentiment categories such as positive, negative, or neutral. This technique plays a crucial role in various fields, including social media monitoring, customer feedback analysis, and market research.

Earlier approaches to sentiment analysis relied on lexicon-based methods or traditional machine learning techniques like support vector machines (SVM) and naive Bayes classifiers. While these methods achieved moderate accuracy, they often struggled with nuances such as context, sarcasm, and complex sentence structures.

The emergence of deep learning and transformer-based models has significantly advanced sentiment analysis. Transformer architectures, particularly BERT (Bidirectional Encoder Representations from Transformers) [2], T5 (Text-to-Text Transfer Transformer) [3], and GPT (Generative Pre-trained Transformer), have set new benchmarks in NLP tasks, including sentiment classification. These models utilize self-attention mechanisms and contextual embeddings to better understand the meaning and relationships within text data.

In this paper, we examine the application of transformer models (BERT, T5, and GPT) to sentiment analysis. Section I introduces the concept of sentiment analysis, while Section II reviews several related studies addressing problems similar to those explored in our research. Section III describes the methods employed in our experiments. In Section IV, we present the datasets used. Section V outlines the experimental setup, including two tests conducted as part of our sentiment analysis and the corresponding results. The achieved outcomes are thoroughly discussed and critically compared with findings from related work. Finally, Section VI summarizes our findings and outlines potential directions for future work.

II. RELATED WORK

In paper by Mengi et al. [4] explores the advancements in Natural Language Processing (NLP), focusing on the transformer models T5 and RoBERTa. T5's text-to-text approach and RoBERTa's optimization for large datasets and long sequences are analyzed for their performance in text summarization and sentiment analysis. The models were fine-tuned using news articles, product reviews, and sentiment data, resulting in significant improvements. T5 showed an increase in ROUGE-L scores from 13.6% to 21.5%, while RoBERTa achieved 94.71% accuracy and a 94.6% F1-score in sentiment analysis. These findings demonstrate the models' effectiveness and pave the way for further advancements in NLP applications.

Vasireddy et al. [5] combines the T5 and GPT-4 models to improve sentiment analysis of YouTube video comments, demonstrating a hybrid approach for more accurate and nuanced results. The process begins with the T5 model performing initial sentiment classification, followed by refinement and contextual validation using GPT-4. This two-step methodology enhances both the speed and depth of analysis, offering detailed insights into user sentiment. The final application provides a web-based tool that allows users to input YouTube URLs and receive comprehensive sentiment analysis. The integration of these advanced models highlights their potential for analyzing large-scale textual data, improving accuracy and offering deeper understanding of public opinion across web media.

In paper by Pipalia et al. [6] explores the effectiveness of pre-trained transformer-based models for sentiment analysis, focusing on models like BERT, XLNet, RoBERTa, DistilBERT, T5, and Bidirectional LSTM. The study compares the performance of these models on the IMDB review dataset, with XLNet achieving the highest accuracy at 96.2%. Key advancements such as the bidirectional nature of transformers and autoregressive techniques in XLNet are highlighted for their superior handling of context and accuracy. The paper also discusses improvements in model efficiency, including the use of DistilBERT, which, while faster, maintains high accuracy. Overall, the research underscores the growing importance of transformer models in NLP tasks, demonstrating their superiority over traditional models like LSTM in various benchmarks.

Liu et al. [7] introduces OpinionSpanT5, a model designed to improve Aspect Sentiment Quad Prediction (ASQP) by addressing the challenge of opinion item extraction in sentiment analysis. Unlike traditional Seq2Seq-based approaches that struggle with missing opinion spans, OpinionSpanT5 incorporates a BiLSTM-CRF layer to first extract opinion items before generating sentiment interpretations using the T5 model. Experimental results on REST15 and REST16 datasets demonstrate that OpinionSpanT5 outperforms existing methods, including Hybrid Graph Convolutional Network BERT (HGCN_BERT) and Generative Aspect-based Sentiment analysis (GAS), achieving the highest precision, recall, and F1 scores. Specifically, OpinionSpanT5 attained an F1-score of 47.47% on REST15 (compared to GAS: 45.98%, HGCN_BERT: 22.15%) and 58.37% on REST16 (compared to GAS: 56.04%, HGCN_BERT: 24.68%). The study highlights the limitations of pipeline methods due to error accumulation and identifies synonym confusion and aspect classification challenges as key areas for future improvement. Despite progress, ASQP remains a challenging task with significant room for advancement.

In paper by Rusnachenko et al. [8] explores the use of decoder-based generative transformers for sentiment analysis of named entities in Russian news articles, focusing on the RuSentNE-2023 dataset. The research evaluates both zero-shot capabilities of large language models (LLMs) and fine-tuning of Flan-T5 using the Chain-of-Thought (CoT) three-hop reasoning framework (THoR). Findings show that zero-shot LLMs perform comparably to fine-tuned BERT-based baselines, while fine-tuning Flan-T5 with THoR improves performance by at least 5%. The best results were achieved with fine-tuned Flan-T5xl, surpassing previous state-of-the-art classifiers. Performance varied based on language, with LLMs generally performing better on English-translated data than original Russian texts. OpenAI's GPT-4 achieved the highest F1-score (54.53%) on translated data, while Mistral-Instruct-7Bv0.2 demonstrated strong multilingual capabilities. Error analysis identified challenges such as incorrect sentiment attribution in multi-entity sentences, confusion between sentiment toward individuals versus events, and difficulty distinguishing sentiment directed at out-of-sentence entities. Future research

will focus on improving reasoning techniques and parameter-efficient tuning for larger models.

III. APPLIED METHODS

The initial step involves data acquisition, for instance, from a social media platform, followed by preprocessing to ensure data quality and consistency. Subsequently, the data is annotated into multiple categories and balanced to maintain an equal distribution of comments across all categories. Once the dataset is prepared, it is utilized for training a language model. The training phase is followed by model validation to assess its performance and generalization capabilities. Finally, the model undergoes testing, and in the last step, the obtained results are analyzed and evaluated (see Figure 1).

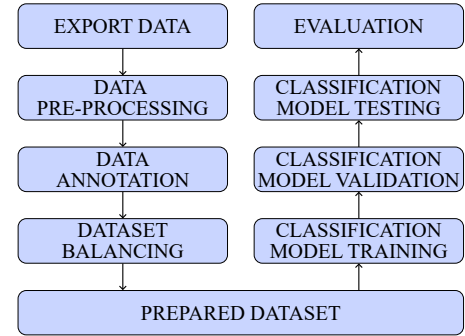


Fig. 1. The entire process from creating a dataset to evaluating the results achieved.

In our experiments, we applied several methods to assess their performance across various tasks. The first approach involved pre-trained BERT models, specifically DistilBERT [9], [10] and RoBERTa [11], [12], which are widely used for natural language processing tasks. For comparison, we also included pre-trained T5 models, particularly mT5 [13], [14] and byT5 [15], [16], known for their versatility in multilingual tasks. The final method we explored involved the use of the large pre-trained language model GPT [17], [18]. In the experiments, we fine-tuned the selected methods on the chosen datasets. To evaluate the results, we used well-known metrics: accuracy, F1-score, precision, and recall.

IV. DATASETS

In our experimental solution, we used three different datasets. The first is the **SentiSK** dataset [19], which we created as part of research focused on sentiment analysis in the Slovak language [20]. SentiSK contains 34 006 comments from the social network Facebook. The comments were obtained using a Python tool for extracting data from web pages, specifically comments under posts by three Slovak politicians. The data preprocessing involved cleaning the text of unwanted characters, including the removal of empty lines, unnecessary spaces, and dots, among others. We used the NLTK library for preprocessing. The dataset was annotated using the annotation tool Prodigy, provided by the Department of Electronics and Multimedia Communications (DEMC). The SentiSK dataset

was annotated into three sentiment categories: 20 668 negative comments, 9 581 neutral comments, and 3 779 positive comments. The distribution of comments across these categories clearly shows that the SentiSK corpus is class-imbalanced. Since the data was taken from posts by Slovak politicians, it contained a large number of negative comments.

The second dataset used in our experimental solution was **the Sentigrade** dataset by Krchnavy et al. [21], which is in Slovak and available online. Sentigrade is a project that resulted from the collaboration between the Faculty of Informatics and Information Technologies at STU BA and the PR agency Seesame. Its creators developed and utilized the dataset for a task involving multi-level text preprocessing to address the complexities of user-generated content on social networks. The original dataset was divided into five groups. For easier comparison with our dataset, we merged the "strongly negative" category with the "negative" category and the "strongly positive" category with the "positive" category. The "neutral" category remained unchanged. In its final structure, the dataset contained a total of 1 588 comments, with 709 labeled as "negative," 573 as "positive," and 306 as "neutral." The Sentigrade dataset was obtained from the Facebook pages of the Seesame company and related comments on these pages.

The third dataset in this experiment was **the Slovak dataset for SA** dataset, created by the authors Machová et al. [22] during their research. The corpus is available online on the website [23]. The Slovak dataset for SA dataset consists of reviews downloaded from e-commerce websites and contains 2 669 negative and 2 573 positive comments.

V. EXPERIMENTS

As part of our research, we focused on two experiments, which are described in Subsection V-A and V-B. In the next step, it was necessary to define the data split for the training-validation-testing process. The data splitting in this experimental setup is implemented using the *train_test_split* function from the *sklearn.model_selection* library. This function is used to split two datasets: *processed_features*, representing input features, and labels, representing the target variables. The parameter *test_size* = 0.2 specifies that 20% of the total data is allocated for the test set. The number of epochs was set to 3 for these experiments. All models used in these experiments are available on GitHub [24].

All experiments were conducted on a DGX server (Technical University of Košice) with the following configuration: 256-core CPU, 1.0 TiB RAM, and 8× NVIDIA A100-SXM4-40GB GPUs (40GB per GPU), running Ubuntu 24.04.2 LTS. SLURM was used for resource management and job scheduling. The software environment was set up using Miniconda with support for PyTorch.

A. Three-Class Classification

In the first experiment, we focused on three-class sentiment classification (positive, neutral, and negative) using two Slovak-language datasets: **SentiSK** and **Sentigrade**. Table I

presents the performance of five Transformer-based models—RoBERTa, DistilBERT, mT5, byT5, and GPT—evaluated across multiple metrics, including accuracy, F1-score, precision, and recall. This experiment allows us to assess the ability of these models to distinguish between a broader spectrum of sentiment expressions in Slovak text.

TABLE I
OVERALL RESULTS OF THE EXPERIMENT FOR THE SentiSK AND SENTIGRADE DATASETS WITH THREE-CLASS CLASSIFICATION.

Corpora	Metric (%)	RoBERTa	DistilBERT	mT5	byT5	GPT
SentiSK	Accuracy	45,38	65,81	63,14	60,02	64,56
	F1-score	47,16	56,44	58,50	50,91	64,20
	Precision	55,03	57,97	58,86	51,89	63,92
	Recall	45,38	55,47	63,14	50,16	64,56
Sentigrade	Accuracy	48,90	60,50	65,89	44,97	55,46
	F1-score	45,33	50,82	56,88	26,16	57,64
	Precision	46,62	53,00	55,42	28,00	68,95
	Recall	48,90	51,47	62,03	34,28	55,46

In the first experiment, we compared the performance of several transformer-based models for the task of three-class sentiment classification on Slovak language datasets. The results show a noticeable variability in performance depending on the model architecture and the dataset used.

On the **SentiSK** dataset, which is large but heavily imbalanced in favor of negative comments, the **GPT** model achieved the best results across all metrics—highest accuracy (64.56%), F1-score (64.20%), precision (63.92%), and recall (64.56%). This performance aligns with the findings of Vasireddy et al. [5], where GPT-4 in combination with a T5 model showed strong results in sentiment classification tasks due to its nuanced contextual understanding. Although we did not use a model ensemble, GPT's generative nature and advanced language comprehension likely contributed to improved classification, especially of less frequent classes.

On the smaller and more balanced **Sentigrade** dataset, the best performance was achieved by the **mT5** model, which obtained the highest accuracy (65.89%) and F1-score (56.88%). These findings support prior work by Mengi et al. [4] and Pipalia et al. [6], which emphasized the strength of T5-based models in text-to-text tasks, particularly when processing shorter or more structured user-generated content. The multilingual nature of mT5 may also have aided its adaptation to Slovak-language inputs.

In contrast, the **RoBERTa** model demonstrated weaker performance across both datasets, with an especially low F1-score of 47.16% on SentiSK. This may be attributed to RoBERTa's pretraining on English-only corpora and its encoder-only architecture, which might be less effective for generative tasks or for managing highly imbalanced class distributions. Similar limitations were observed in Liu et al. [7], where generative approaches such as OpinionSpanT5 outperformed traditional BERT architectures in more complex sentiment scenarios.

DistilBERT achieved consistently moderate results. Given its lower parameter count and faster inference time, it remains a practical option for applications where computational efficiency is a priority, as also supported by Pipalia et al. [6].

Interestingly, the **byT5** model performed notably poorly

on the Sentigrade dataset, with an F1-score of just 26.16%. This suggests that byte-level encoding may be less effective on small, linguistically homogeneous datasets and might require larger or more diverse input data to perform well.

It is important to note that the performance of models on the **SentiSK** dataset may have been influenced by the strong class imbalance—particularly the overrepresentation of negative comments. This factor can negatively affect the F1-score and precision for minority classes (e.g., positive comments), which may explain the lower performance observed in some models.

Overall, our results confirm that model architecture, dataset size and balance, and language-specific adaptation play a critical role in sentiment classification performance for low-resource languages such as Slovak. These insights provide a practical overview of suitable model choices for varying data conditions and can serve as a basis for future development of Slovak sentiment classifiers.

B. Two-Class Classification

In the second experiment, we focused on binary sentiment classification (positive vs. negative) using three Slovak-language datasets: **SentiSK**, **Sentigrade**, and the **Slovak dataset for SA**. Table II presents the performance of five Transformer-based models across these corpora, evaluated using accuracy, F1-score, precision, and recall.

The results in Table II showcase the performance of Transformer-based models—RoBERTa, DistilBERT, mT5, byT5, and GPT—on three different Slovak sentiment analysis datasets: SentiSK, Sentigrade, and Slovak dataset for SA. The models were evaluated using accuracy, F1-score, precision, and recall.

TABLE II
OVERALL RESULTS OF THE EXPERIMENT FOR THE SentiSK,
SENTIGRADE AND SLOVAK DATASET FOR SK CORPORA WITH TWO-CLASS
CLASSIFICATION.

Corpora	Metric (%)	RoBERTa	DistilBERT	mT5	byT5	GPT
SentiSK	Accuracy	70.30	87.43	89.23	86.91	87.32
	F1-score	70.45	86.89	88.04	68.18	87.00
	Precision	70.71	86.58	88.05	76.97	86.76
	Recall	70.30	87.43	89.15	64.86	87.32
Sentigrade	Accuracy	87.94	66.15	81.98	45.14	86.46
	F1-score	87.94	65.46	81.96	45.11	86.64
	Precision	88.00	71.32	81.96	45.84	86.51
	Recall	87.94	68.35	81.98	45.89	86.46
Slovak dataset for SA	Accuracy	92.62	90.08	84.23	52.91	93.38
	F1-score	92.62	90.07	81.16	43.97	93.38
	Precision	92.65	90.17	83.14	60.97	93.40
	Recall	92.62	90.08	84.22	53.65	93.38

Across all datasets, the **GPT** model achieved top results or performed very competitively. Notably, it achieved the highest performance on the **Slovak dataset for SA**, reaching an F1-score and accuracy of 93.38%. These results align with the findings of Vasireddy et al. [5], who highlighted GPT’s ability to understand sentiment nuances in text due to its generative nature and advanced contextual understanding. Its robustness across datasets of different sizes and characteristics makes it a reliable choice for sentiment analysis in low-resource languages.

The **mT5** model, although slightly behind GPT on the Slovak dataset for SA, achieved the best overall performance on the **SentiSK** dataset, with an accuracy of 89.23% and an F1-score of 88.04%. This supports the observations from Mengi et al. [4] and Pipalia et al. [6], who noted mT5’s strong performance in multilingual text generation and classification tasks.

Surprisingly, **RoBERTa** performed best on the **Sentigrade** dataset, achieving perfect alignment between accuracy and F1-score (87.94%). Although pretrained on English data, RoBERTa’s performance on this small and more balanced dataset suggests that its encoder architecture can still generalize well under certain conditions, possibly due to the simpler binary classification task. However, its performance dropped significantly on the more imbalanced or linguistically varied datasets like SentiSK, consistent with prior findings in Liu et al. [7].

DistilBERT delivered consistent and solid results across all datasets, particularly on the SentiSK (F1-score of 86.89%) and Slovak dataset for SA (90.07%). Its lightweight architecture and faster inference make it suitable for practical applications where computational resources are limited, echoing conclusions from Pipalia et al. [6].

The **byT5** model exhibited the weakest performance overall, particularly on the Sentigrade and Slovak SA datasets. With an F1-score of only 43.97% on the Slovak dataset for SA and 45.11% on Sentigrade, it lagged significantly behind other models. This suggests that byte-level tokenization might not be effective for small or domain-specific datasets and may require more data to leverage its full potential.

It is important to note that dataset characteristics likely influenced the results. For example, the class imbalance in **SentiSK** and the relatively small size of **Sentigrade** may have affected model generalization. These factors highlight the need for careful dataset preparation and class balancing, especially when working with low-resource languages such as Slovak.

In conclusion, the comparative analysis confirms that generative and multilingual models like GPT and mT5 are well-suited for sentiment classification in Slovak, while distilled and encoder-based models offer a trade-off between performance and efficiency. The findings also reinforce that dataset quality, size, and balance significantly impact model outcomes and should be considered in future research and application development.

VI. CONCLUSION

In this paper, we explored the application of transformer-based models—BERT, T5, and GPT—in sentiment analysis, specifically for the Slovak language. Our study focused on evaluating these models on three distinct datasets: SentiSK, Sentigrade, and the Slovak dataset for SA. We conducted both three-class and two-class classification experiments, analyzing the effectiveness of each model using well-established metrics such as accuracy, F1-score, precision, and recall.

Our findings indicate that transformer models significantly enhance sentiment classification performance compared to

traditional methods. GPT and mT5 emerged as the most effective models across different datasets, demonstrating strong generalization capabilities and high classification accuracy. In the three-class classification task, DistilBERT performed best on SentiSK, while mT5 showed the highest accuracy on Sentigrade. GPT consistently performed well, particularly excelling in F1-score and precision.

In the two-class classification task, mT5 achieved the highest accuracy on SentiSK, while RoBERTa performed best on Sentigrade, and GPT outperformed all other models on the Slovak dataset for SA. These results highlight the importance of dataset-specific model selection, as performance varied depending on the nature and characteristics of the datasets.

Despite the promising results, challenges remain, particularly in dealing with class-imbalanced datasets and nuanced language expressions such as sarcasm and irony. Future research should explore additional fine-tuning strategies, data augmentation techniques, and hybrid approaches combining multiple models to further enhance sentiment classification accuracy. Additionally, expanding the scope of sentiment analysis to more diverse domains, including spoken language and multimodal data, represents a promising direction for further advancements in this field.

Overall, our study contributes to the ongoing research in sentiment analysis by demonstrating the effectiveness of modern transformer models for Slovak text classification. In addition, it extends our previous work [20] by incorporating a broader range of models, including RoBERTa, DistilBERT, mT5, byT5, and GPT. The insights gained from this work can aid in the development of more robust NLP applications tailored for sentiment analysis in underrepresented languages.

APPENDIX

In this Section, we have compiled a comprehensive list of abbreviations used throughout this paper, arranged alphabetically for ease of reference:

- AI: Artificial Intelligence
- API: Application Programming Interface
- ASQP: Aspect Sentiment Quad Prediction
- BERT: Bidirectional Encoder Representations from Transformers
- BiLSTM: Bidirectional Long Short-Term Memory
- byT5: Byte-to-Byte Text-to-Text Transfer Transformer
- CoT: Chain-of-Thought
- CRF: Conditional Random Fields
- DEMC: Department of Electronics and Multimedia Communications
- GAS: Generative Aspect-based Sentiment analysis
- GPT: Generative Pre-trained Transformer
- GPT-4: Generative Pre-trained Transformer 4
- HGCN: Hybrid Graph Convolutional Network
- IMDB: The Internet Movie Database
- LM: Language Model
- LLMs: Large Language Models
- LSTM: Long Short-Term Memory
- mT5: multilingual Text-to-Text Transfer Transformer
- NLP: Natural Language Processing
- RoBERTa: Robustly Optimized BERT Pretraining Approach
- SA: Sentiment Analysis
- Seq2Seq: Sequence-to-sequence
- STU: Slovak University of Technology in Bratislava
- SVM: Support Vector Machines
- T5: Text-to-Text Transfer Transformer
- THoR: Three-Hop Reasoning framework
- URL: Uniform Resource Locator
- XLNet: eXtreme Language Net

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REFERENCES

- [1] T. Shaik, X. Tao, C. Dann, H. Xie, Y. Li, and L. Galligan, "Sentiment analysis and opinion mining on educational data: A survey," *Natural Language Processing Journal*, vol. 2, p. 100003, 2023.
- [2] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, 2019, pp. 4171–4186.
- [3] C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu, "Exploring the limits of transfer learning with a unified text-to-text transformer," *The Journal of Machine Learning Research*, vol. 21, no. 1, pp. 5485–5551, 2020.
- [4] R. Mengi, H. Ghorpade, and A. Kakade, "Fine-tuning t5 and roberta models for enhanced text summarization and sentiment analysis," *The Great Lakes Botanist*, vol. 12, 2023.
- [5] I. Vasireddy, B. Sriveni, K. Prathyusha, and A. Kandi, "Sentiment analysis of web media using hybrid model with t5 and gpt-4 models," in *2024 2nd International Conference on Recent Trends in Microelectronics, Automation, Computing and Communications Systems (ICMACC)*. IEEE, 2024, pp. 634–639.
- [6] K. Pipalia, R. Bhadja, and M. Shukla, "Comparative analysis of different transformer based architectures used in sentiment analysis," in *2020 9th international conference system modeling and advancement in research trends (SMART)*. IEEE, 2020, pp. 411–415.
- [7] P. Liu and W. Guo, "A sentiment analysis model based on text generation-opinionspant5," in *2024 IEEE 2nd International Conference on Control, Electronics and Computer Technology (ICCECT)*. IEEE, 2024, pp. 1536–1540.
- [8] N. Rusnachenko, A. Golubev, and N. Loukachevitch, "Large language models in targeted sentiment analysis for russian," *Lobachevskii Journal of Mathematics*, vol. 45, no. 7, pp. 3148–3158, 2024.
- [9] H. Face, "Distilbert-distilbert-base-multilingual-cased," <https://huggingface.co/distilbert/distilbert-base-multilingual-cased>, Accessed 2024.
- [10] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter," *ArXiv*, vol. abs/1910.01108, 2019.
- [11] H. Face, "Facebookai-roberta-base," <https://huggingface.co/FacebookAI/roberta-base>, Accessed 2024.
- [12] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, "Roberta: A robustly optimized BERT pretraining approach," *CoRR*, vol. abs/1907.11692, 2019. [Online]. Available: <http://arxiv.org/abs/1907.11692>
- [13] H. Face, "Google-mt5-small," <https://huggingface.co/google/mt5-small>, Accessed 2024.
- [14] L. Xue, N. Constant, A. Roberts, M. Kale, R. Al-Rfou, A. Siddhant, A. Barua, and C. Raffel, "mt5: A massively multilingual pre-trained text-to-text transformer," *arXiv preprint arXiv:2010.11934*, 2020.
- [15] H. Face, "Google-byt5-small," <https://huggingface.co/google/byt5-small>, Accessed 2024.
- [16] L. Xue, A. Barua, N. Constant, R. Al-Rfou, S. Narang, M. Kale, A. Roberts, and C. Raffel, "Byt5: Towards a token-free future with pre-trained byte-to-byte models," *Transactions of the Association for Computational Linguistics*, vol. 10, pp. 291–306, 2022.

- [17] M. Kondela, "Slovak GPT-J-405M," <https://huggingface.co/Milos/slovak-gpt-j-405M>, 2022.
- [18] B. Wang, "Mesh-Transformer-JAX: Model-Parallel Implementation of Transformer Language Model with JAX," <https://github.com/kingoflolz/mesh-transformer-jax>, May 2021.
- [19] Z. Sokolová, "Sentisk dataset," <https://huggingface.co/datasets/TUKE-KEMT/senti-sk>, Accessed 2025.
- [20] Z. Sokolová, M. Harahus, J. Juhár, M. Pleva, J. Staš, and D. Hládek, "Comparison of machine learning approaches for sentiment analysis in slovak," *Electronics*, vol. 13, no. 4, p. 703, 2024.
- [21] R. Krchnavy and M. Simko, "Sentiment analysis of social network posts in slovak language," in *2017 12th International Workshop on Semantic and Social Media Adaptation and Personalization (SMAP)*. IEEE, 2017, pp. 20–25.
- [22] K. Machová, M. Mikula, X. Gao, and M. Mach, "Lexicon-based sentiment analysis using the particle swarm optimization," *Electronics*, vol. 9, no. 8, p. 1317, 2020.
- [23] K. Machová, "Slovak dataset for sentiment analysis," <https://kristina.machova.website.tuke.sk/useful/>, Accessed 2023.
- [24] Z. Sokolová, "Used models," https://github.com/tpackolus/Radio_sentiment/tree/main, Accessed 2025.